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Demonstration question paper generated from available previous papers - for practice only

B.Tech. I Semester Examination, 2025

Regulation: R20 | University: JNTUK

Artificial Intelligence (21CS53)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Introduction & Intelligent Agents]

- a) Explain the various types of agents with examples.
- b) Solve the given problem using A* search and explain admissibility.

UNIT - 2 [Adversarial Search & Constraint Satisfaction]

- a) Explain the minimax algorithm with alpha beta pruning.
- b) Formulate the given problem as a CSP and solve it.

UNIT - 3 [Knowledge Representation & Planning]

- a) Convert the given statements into first order logic.
- b) Explain resolution refutation with an example.

UNIT - 4 [Reasoning Under Uncertainty]

- a) Explain the working of a Bayesian network with an example.
- b) Explain the ID3 decision tree algorithm.

UNIT - 5 [Machine Learning & NLP Basics]

- a) Write short notes on expert systems.
- b) Write short notes on expert systems.

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